

Project Proposal: GPU-Accelerated Agent-Based Simulation for Cell Dynamics

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Introduction

Agent-based modeling (ABM) is a widely used computational approach for simulating complex biological systems, particularly in cancer research. By representing individual cells as agents with distinct behaviors, ABMs can model processes such as tumor growth, immune responses, and viral therapy interactions. However, existing models often suffer from computational inefficiencies and simplified biological assumptions, limiting their scalability and accuracy. Traditional ABMs typically execute agent updates sequentially, making them computationally expensive for large-scale simulations. Additionally, many models assume synchronous updates, which do not reflect the asynchronous nature of real biological interactions.

This project aims to complete and optimize a GPU-accelerated 2D agent-based simulation for studying cancer growth, immune response, and engineered viral therapy. By implementing PyOpenCL-based GPU acceleration, the model will improve performance and scalability, enabling real-time simulations with large populations of interacting agents. Additionally, a modular simulation engine will be developed to ensure that different biological processes, such as chemokine diffusion and cell interactions, operate in parallel, rather than being bottlenecked by the slowest process. While the primary focus is on biological accuracy and computational efficiency, the project will also explore machine learning techniques such as RiTINI and MIOFlow to analyze cell-cell interactions and predict cancer-immune-virus progression.

Background / Related Works

Computational modeling plays an essential role in cancer biology, particularly in understanding how tumors interact with the immune system and therapeutic interventions. The Krishnaswamy Lab has developed several data-driven computational methods, including RiTINI, which infers regulatory networks from dynamic cell-cell interactions, and MIOFlow, which uses optimal transport to reconstruct continuous cell-state transitions. These methods have been applied to spatial transcriptomics and single-cell RNA sequencing (scRNA-seq) data, offering insights into tumor microenvironments and immune cell behavior. Integrating these techniques into agent-based models could enhance their biological interpretability and predictive power.

In addition to improving biological accuracy, this project addresses a fundamental computational challenge in agent-based modeling: parallelization and performance scaling. GPU acceleration has been successfully applied in epidemiological models and fluid dynamics simulations, but its application to cell-scale cancer modeling remains limited. Implementing PyOpenCL-based parallel processing will enable real-time simulation of complex cell interactions, making it possible to study large-scale tumor-immune-virus dynamics efficiently. By leveraging both

biological modeling techniques and high-performance computing, this project aims to create a scalable, modular framework for studying engineered viral therapies in cancer treatment.

Problem Description

This project addresses the following research question:

How can an engineered virus effectively target and eliminate cancer cells while avoiding excessive collateral damage to healthy cells?

To answer this, the model will:

1. Simulate cancer growth in a 2D cellular environment.
2. Model engineered viral infection dynamics and its interactions with tumor cells.
3. Incorporate immune response mechanisms (e.g., CD4+ and CD8+ T cell activation).
4. Improve biological accuracy by refining chemokine diffusion, immune cell recruitment, and tumor-virus interactions.
5. Implement GPU acceleration (PyOpenCL) to allow parallel agent updates and improve computational efficiency.
6. Develop a modular simulation engine where independent biological processes run asynchronously, preventing slowdowns.
7. Test scalability and performance by comparing GPU vs. CPU execution times and analyzing large-scale tumor-immune interactions.
8. (Secondary Goal) Apply machine learning techniques such as RiTINI and MIOFlow to analyze simulation outputs and predict disease progression.

This project provides a scalable and high-performance computational framework for studying cancer-immune-virus interactions and testing oncolytic viral therapies.

Planned Approaches

1. Completing the 2D Agent-Based Model (Weeks 1–6)
 - a. Implement missing agent behaviors, ensuring proper cancer proliferation, immune cell recruitment, and viral spread dynamics.
 - b. Improve chemokine diffusion modeling, ensuring immune cells are attracted to infection sites in a biologically accurate way.
 - c. Refine rules for immune cell activation and tumor-virus interactions.
2. Implementing GPU Acceleration (Weeks 6–10)
 - d. Parallelize agent updates using PyOpenCL, enabling cells, viruses, and chemokines to update simultaneously.
 - e. Optimize chemokine diffusion as a continuous GPU process.
 - f. Benchmark performance improvements, measuring speedup vs. CPU-based execution.
3. Machine Learning Analysis & Perturbation Experiments (Weeks 11–14)
 - g. (Secondary) Apply RiTINI to extract interaction networks from the simulation data.

- h. (Secondary) Use MIOFlow to reconstruct cell-state trajectories over time.
- i. Run perturbation experiments to analyze how different viral therapy strategies affect tumor growth and immune response.

Schedules / Timelines

Week	Milestone
1–2	Implement missing agent behaviors (cancer, immune, viral)
3–5	Finalize chemokine diffusion, immune activation, and infection rules
6–8	Implement PyOpenCL-based GPU acceleration for agent updates
9–10	Optimize and benchmark performance improvements
11–12	Apply RiTINI for network inference & MIOFlow for trajectory modeling
13–14	Conduct perturbation experiments & analyze results

List of Deliverables

- A 2D agent-based simulation modeling cancer proliferation, immune response, and viral therapy.
- A GPU-accelerated simulation engine using PyOpenCL for parallel agent/environment updates.
 - Performance benchmarking and scalability analysis comparing CPU vs. GPU execution speed.
- Improved biological accuracy through refined chemokine diffusion, immune recruitment, and tumor-virus interactions.
- (Secondary) Graph-based and trajectory learning analysis using RiTINI for regulatory network inference and MIOFlow for predicting cancer-immune-virus progression.
- A final research report and presentation summarizing methodology, results, and future applications.

Future Work

- Extend the model to include 3D tissue dynamics.
- Explore evolutionary optimization of viral therapies.
- Develop interactive visualizations of the simulation for educational purposes.

References

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Next Steps

- Confirm key simulation parameters (e.g., viral infection rates, immune cell behavior).
- Define data sources (synthetic vs. real single-cell transcriptomics datasets).
- Implement baseline simulation and validate against experimental data.